Experiment 5

Fourier Series Analysis of Periodic Signals

OBJECTIVE 1:

To analyze Fourier series of continuous-time periodic signals and analyze their waveform characteristics.

Procedure

1. Define the time vector and fundamental frequency.

2. Generate the half-wave rectified sinusoidal periodic signal.

3. Compute the Fourier coefficients up to N = 10.

4. Construct the Fourier-series approximation.

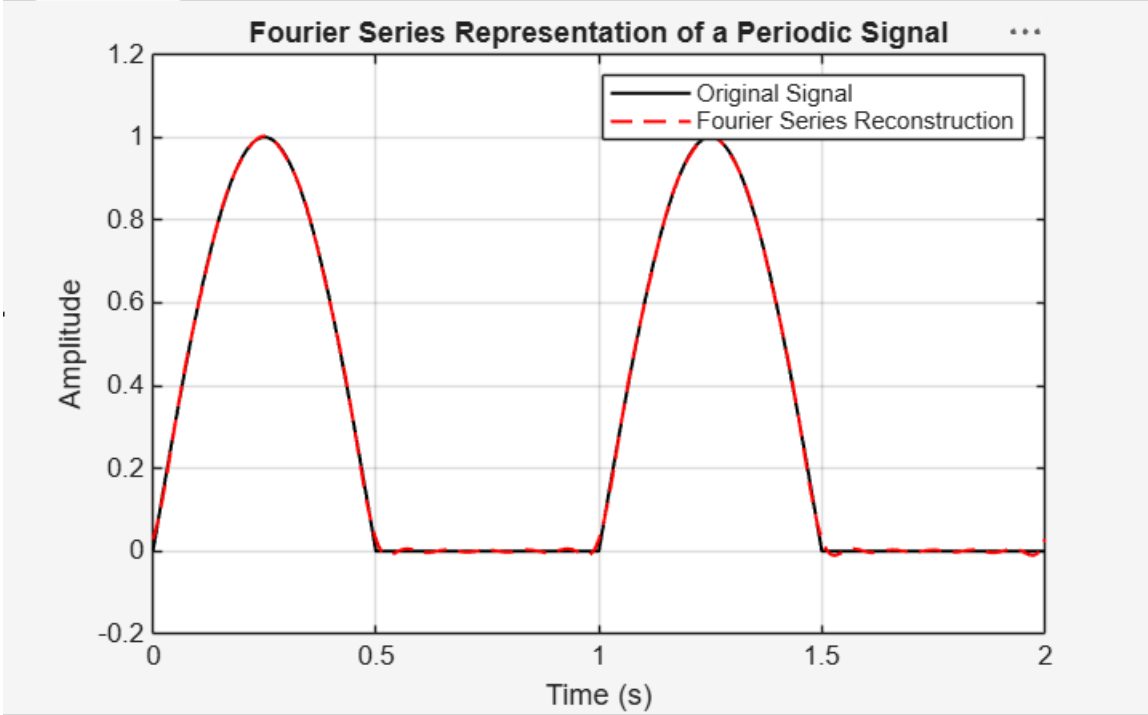
5. Plot the original and reconstructed signals and compare them.

MATLAB Coding

```
clc; clear; close all;
%% 5(i) Analysis of Fourier Series Representation of Continuous-Time Periodic Signals
figure;
% Fourier Series Representation of a Periodic Signal
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
```

```
an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
endxFS = a0*ones(size(t));
for k = 1:N
xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
+ bn(k)*sin(2*pi*k*f0*t);
end
plot(t,x,'k','LineWidth',1.2); hold on;
plot(t,xFS,'r--','LineWidth',1.2);
grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Fourier Series Representation of a Periodic Signal');
legend('Original Signal','Fourier Series Reconstruction');
```

OUTPUT:



RESULT:

OBJECTIVE 2:

To reconstruct periodic signals using Fourier series and verify the reconstructed waveform with the original signal

PROCEDURE:

1. Generate the periodic signal over two periods.

2. Compute Fourier coefficients from one period.

3. Construct the reconstructed signal using 10 harmonics.

4. Calculate the error signal.

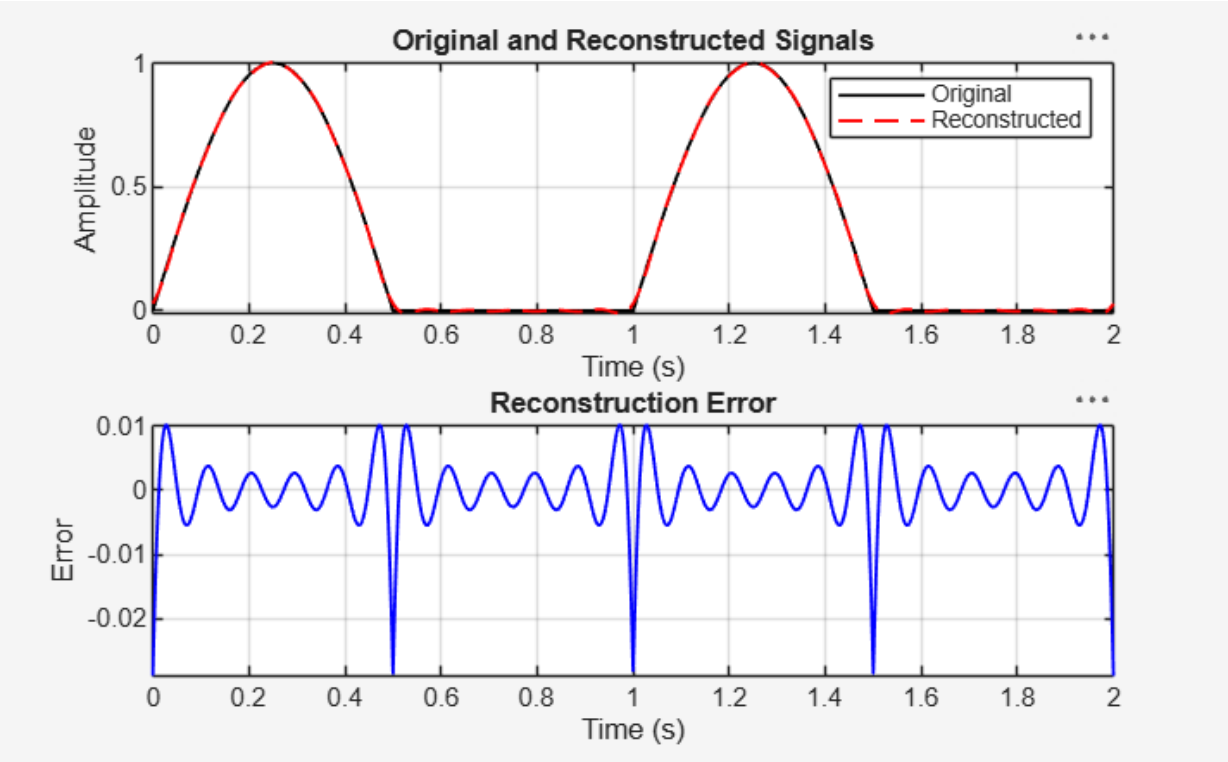
5. Plot the original, reconstructed and error signals.

MATLAB CODE:

```
clc; clear; close all;
%% 5(iv) Verification of Fourier Series Reconstruction and Analysis of Signal Characteristics
figure;
% Fourier Series Reconstruction and Error Analysis
f0 = 1;
T = 1/f0;
t = linspace(0,2*T,4000);
x = max(sin(2*pi*f0*t),0);
N = 10;
a0 = (2/T)*trapz(t(1:2000),x(1:2000));
an = zeros(1,N); bn = zeros(1,N);
tt = t(1:2000); xx = x(1:2000);
for k = 1:N
an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));end
xr = a0/2*ones(size(t));
for k = 1:N
```

```
xr = xr + an(k)*cos(2*pi*k*f0*t) + bn(k)*sin(2*pi*k*f0*t);
end
err = x-xr;
subplot(2,1,1);
plot(t,x,'k',t,xr,'r--','LineWidth',1.1);
grid on; xlabel('Time (s)'); ylabel('Amplitude');
title('Original and Reconstructed Signals');
legend('Original','Reconstructed');
subplot(2,1,2);
plot(t,err,'b'); grid on;
xlabel('Time (s)'); ylabel('Error');
title('Reconstruction Error');
```

OUTPUT:



RESULT: